

Setup / hold

A flip-flop does not sample D at an instant, it needs setup time before the clock edge and hold time after

Windows, margins, Q delay

- Starter case: clean pass with comfortable margins
- D settles at time fourteen
- Data stays low through the hold window after the edge
- Q moves at edge plus t_{cq}
- Drag D later into the red setup band and setup fails
- Flip D too soon after the edge and hold fails

Browser lab

Digital Design and Verification Platform

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Tools

INTERACTIVE TOOL

Setup / hold explainer

See why data must be stable **before** (setup) and **after** (hold) the capturing clock edge. Drag the data transition, tweak delays, and watch the annotated windows go green or red. Starter: a clean pass with comfortable margins.

Starter example: clean setup/hold — data settles before the green tsu window and stays through th. Drag the D transition or raise tsu to create a violation.

Load starter example

Challenges 0/22

Pick one

Quiz: setup

Setup time is the requirement that data be stable...

☐ for at least tsu before the capturing clock edge

☐ only after the clock edge

☐ only during reset

☐ never — clocks are async

Start

Show hint

Check

Next

Stop

Idle

Setup hold starter

learn_digital — setup hold

Workbook practice

- In the workbook track, draw one posedge with t_{su} and t_h bands on the D line
- For edge at twenty, t_{su} two, D change at nineteen, does setup pass?
- For hold t_h one, D change at twenty point five, does hold pass?
- Sketch a two-FF path and label t_{cq} , t_{pd} , and t_{su}
- Name one pitfall: changing D on the same cycle you expect Q to capture it

Pitfalls to watch

- Do not confuse this teaching diagram with full static timing analysis or SPICE
- Setup and hold are flip-flop requirements, combo logic delay is separate
- And remember: the browser lab is literacy
- Real chips still need SDC constraints, corners

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, mark setup and hold on one wave
- When you are ready, take the short quiz, then continue to reset timelines